

Report

Group 3

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1. Project Summary using the CRISP-DM Process

Business Understanding

The goal of the project was to predict customer churn in the telecommunications industry. Customer churn occurs when customers cancel their subscription with a service provider. Telecom markets are highly competitive and switching costs for customers are relatively low, meaning customers can easily change providers. Churn has a direct impact on company revenue because losing customers reduces recurring income and increases marketing costs needed to acquire new customers. Research shows that even small improvements in customer retention can significantly increase profits. Therefore, identifying customers who are likely to churn enables companies to take proactive actions such as targeted offers or customer support interventions. The objective of the project was to develop a predictive model that identifies customers at risk of churn so telecom companies can implement retention strategies before customers leave.

Data Understanding

The project used the Telco Customer Churn dataset, which contains customer information from a telecommunications company. The dataset originally included 7,043 customers and 21 variables, and after cleaning 7,032 observations remained. The target variable was Churn, a binary variable indicating whether a customer left the service. About 26.5% of customers churned, showing moderate class imbalance.

The dataset includes several types of variables:

Numerical variables: tenure (months with the company), MonthlyCharges and TotalCharges

Categorical variables: demographic attributes, service features and account information.

Data Preparation

Several preprocessing steps were applied before training the models.

First, the TotalCharges variable was converted to numeric format, and rows containing invalid values were removed. The customerID variable was dropped because it is only an identifier and does not contribute to prediction.

Categorical variables were transformed using one-hot encoding, while numerical variables were standardized using a scaler to ensure comparable value ranges.

Finally, the dataset was divided into training and testing sets using an 80/20 split with stratified sampling, ensuring that the proportion of churn and non-churn customers remained consistent across the datasets.

Modeling

Two predictive models were developed and compared.

Logistic Regression was used as a baseline model because it is simple, interpretable, and commonly used for classification tasks. The model used L2 regularization, a regularization parameter of $C = 1.0$, and a maximum of 1000 iterations.

Gradient Boosted Decision Trees (GBDT) was selected as a more advanced model capable of capturing nonlinear relationships in the data. The configuration included 200 trees, a learning rate of 0.05, and a maximum depth of 3.

Both models were implemented using a preprocessing pipeline to ensure consistent treatment of numerical and categorical variables during training and evaluation.

Evaluation

Model performance was evaluated using several metrics: accuracy, precision, recall, F1-score, and ROC-AUC. Because the dataset contains class imbalance, recall and F1-score were particularly important to assess how well the model identifies churners.

The results showed that:

- Logistic Regression achieved an accuracy of 0.804, recall of 0.572, and F1-score of 0.608.
- Gradient Boosting achieved an accuracy of 0.792, recall of 0.529, and F1-score of 0.576.
- Both models achieved similar ROC-AUC scores (~0.84).

Although Gradient Boosting produced a slightly higher ROC-AUC, Logistic Regression achieved better recall and F1-score. Since identifying churners is critical in this application, Logistic Regression was selected as the preferred model.

The analysis also identified important churn drivers, including month-to-month contracts, higher monthly charges, and fiber internet services, while long-term contracts and longer tenure reduced churn risk.

Deployment

In a real business environment, the churn prediction model could be integrated into the company's CRM system. Each customer would receive a churn risk score based on their characteristics and service usage.

Customers with high predicted churn probability could then be targeted with retention strategies such as discounts, contract upgrades, or proactive customer support.

The model would need to be updated regularly with new customer data, and integration with marketing or customer service systems would allow companies to act on the predictions effectively.

2. Estimated Workload Distribution

Based on the project work, the approximate time distribution across CRISP-DM phases was:

Phase	Estimated Time
Business understanding	15%
Data understanding	20%
Data preparation	30%
Modeling	20%
Evaluation	10%
Deployment considerations	5%

The data preparation phase consumed the largest portion of time, as cleaning, transforming, and encoding the dataset required substantial work.

If we were to repeat the project within the same timeframe, we would likely allocate more time to modeling and evaluation, particularly for hyperparameter tuning and experimenting with additional algorithms. This could potentially improve predictive performance.

3. Main Strengths of the Project

One major strength of the project was the clear business relevance of the problem. Customer churn prediction is widely used in industry and directly supports decision-making.

Another strength was the comparison of two different modeling approaches, which helped demonstrate the trade-off between model complexity and interpretability.

The project also successfully connected technical results to business insights, such as identifying key churn drivers like short contracts and high monthly charges. This type of interpretability is valuable for stakeholders who need to understand why predictions are made.

4. Weak Points and Future Improvements

One weakness of the project was the limited exploration of alternative models and hyperparameter tuning. With more time, additional algorithms such as Random Forests, XGBoost, or neural networks could have been tested.

Another limitation was that the dataset contained only static customer attributes. In real-world scenarios, companies also use behavioral data such as service usage patterns or customer interactions.

In future projects, these limitations could be addressed by:

- performing deeper hyperparameter optimization
- testing additional algorithms
- incorporating time-based behavioral features
- applying uplift modeling to target customers most responsive to retention actions

These improvements could increase both predictive accuracy and the business value of the solution.